

1) Explain the concept of mapping the data onto aesthetics in class visualization.

Mapping data on aesthetics is the process of assigning data values to visual properties (aesthetics) such as position, size, shape and colour. It helps convert raw data into meaningful visualization.

### Visual Aesthetics

- \* position :- Shows where data points are placed
- \* size :- Represents the magnitude of values
- \* shape :- Differentiates categories
- \* colour :- Represents the categories or numerical values.

### Representation of the different data types

1. Categorical data :- consistency of numbers that
2. or represents groups or categories attributes than numerical values. It is used to classify or label items.

eg:- Blood group  
Gender

\* Cardinal data: It consistency of data which contain the ordinary information

Eg:- low medium high

\* Quantitative data:

consistency of size to show numerical values

Eg:- Palle's bars represents higher sales

proper mapping makes data easy to understand and compare.

## ② 2. Role of scales in data visualization

A scale converts data values into visual elements like position size or color

Types of scales.

\* Linear scale: Equal data values have equal spacing

\* Logarithmic scale: used for very large data ranges

\* Ordinal scale: used for ordered categories  
(low, Medium, High)

\* Color scale: uses different colors to represent values

## Importance:

- \* Ensures accurate representation
- \* Makes graphs easy to read
- \* Helps data values in visual elements

## The Role of scales in data

## Impact of inappropriate scaling

- \* Misleading graphs
- \* Hidden trends
- \* Exaggerated differences
- \* Incorrect Interpretation

The scale maps data values to visual properties such as position size or color ensuring accurate representation.

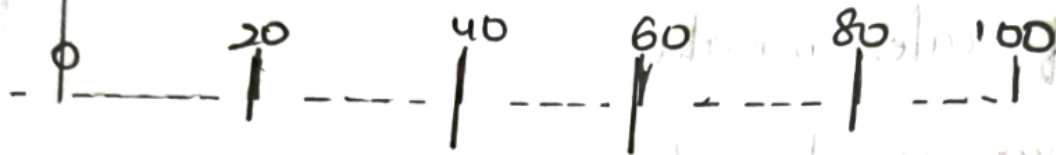
## Conclusion

choosing the correct scale is essential for accurate and effective visualization

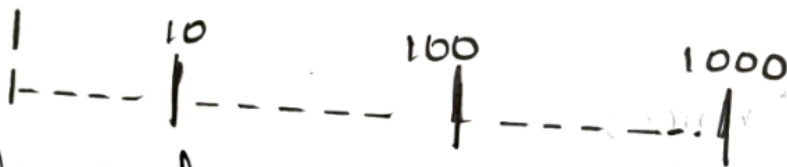
## Applications

- Business : compare sales and profit
- Healthcare : Display patient health data
- Education : show grades and performance
- Weather : Represents temperature using colors and scales.

linear scale



logarithmic scale



colour scale

low temperature      high

Temperature

Blue → Green → yellow → Red

5) Describe the role of color in the data visualization

Introduction  
color is used to organize compare and emphasize information in data visualization

uses of color

1. Distinguishing categories
  - Different colors represents different groups

• Example :: HR (Blue)

## 2. Representing data values

- color gradients show increasing or decreasing values

example :: Blue  $\rightarrow$  cold, Red  $\rightarrow$  Hot

## 3. Highlighting Important Information

- Bright colors draw attention
- Examples: Green for profit

## principles of effective color selection

- Contrast:- Ensure readability
- Accessibility: use colorblind-friendly palettes
- Visual Harmony: use balanced and consistent colors

## Effects of poor color choices

- confusing visualization
- Misinterpretation of data
- Reduced Readability

## conclusion

Effective use of color improves clarity and communication, while poor color choices can mislead users and reduce visualization quality.